

Sounak Mondal

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RESEARCH INTERESTS

I am interested in research on **vision-language modeling**. My PhD thesis focuses on using **vision-language representation learning** and **multimodal foundation models (e.g., Multimodal LLM)** for modeling human visual attention (eye gaze).

EDUCATION

Stony Brook University, Stony Brook, NY

Doctor of Philosophy, Computer Science (**GPA: 4.0/4.0**)

Aug 2020 – June 2026

Advisors: Minh Hoai Nguyen, Dimitris Samaras, Gregory Zelinsky

Jadavpur University, Kolkata, India

Bachelor of Engineering, Computer Science & Engineering (**GPA: 8.65/10, Rank: 3/54**)

Aug 2013 – May 2017

EXPERIENCE

Meta

Research Scientist Intern, Reality Labs Research, Burlingame, CA

Jun 2025 – Dec 2025

- Modeling Vision and Sensor Data via Multimodal LLMs.

Research Scientist Intern, Reality Labs Research, Redmond, WA

Jun 2024 – Dec 2024

- Vision-Language Modeling of Eye Gaze Behavior (published at ICCV 2025, patent filed)

CV Lab, Stony Brook University

Graduate Researcher, Stony Brook, NY

Nov 2020 – Present

- Vision-Language Modeling for gaze prediction (CVPR'23, ECCV'24), and gaze decoding (ICCV'25 accepted + 1 under review)
- Gaze prediction modeling for search (ECCV'22, CVPR'24, CVPR'25); multimodal modeling for gaze estimation (ECCV'24)
- Modeling attentional states, e.g., mind-wandering vs. sustained attention, from eye gaze during online learning (ongoing)

Samsung

NLP Engineer, Natural Language Processing Team, Bangalore, India

Jun 2017 – Aug 2020

- Enhancements of **Bixby** virtual assistant.
- Low resource intent classification via transfer learning
- Sequence labeling models for Named Entity Recognition and speech end-point detection
- Lightweight and fast text classification architecture (ICSC'20)

Summer Intern, SVoice Team, Bangalore, India

May 2016 – Jul 2016

- Context awareness in SVoice platform for Natural Language Processing

UII America

Research Intern, Computer Vision, Cambridge, MA

May 2023 – Aug 2023

- Vision-Language Modeling: Scene Graph Generation from captions using Large Language Models (LLMs).

Indian Statistical Institute

Undergraduate Researcher, Kolkata, India

Jul 2015 – Jun 2017

- Video Action Recognition/Detection (ICAPR'17, ICVGIP Workshop'16).

SELECTED PUBLICATIONS

Gaze-to-text Generation: Beyond Categorical Decoding of Human Attention

ECCV 2026

Sounak Mondal, D. Samaras, G. Zelinsky, M. Hoai

Gaze-Language Alignment for Zero-Shot Prediction of Visual Search Targets from Human Gaze Scanpaths

ICCV 2025

Sounak Mondal, N. Sendhilnathan, T. Zhang, Y. Liu, M. Proulx, M.L. Iuzzolino, C. Qin, T.R. Jonker

Look Hear: Gaze Prediction for Speech-directed Human Attention

ECCV 2024

Sounak Mondal, S. Ahn, Z. Yang, N. Balasubramanian, D. Samaras, G. Zelinsky, M. Hoai

Gazeformer: Scalable, Effective and Fast Prediction of Goal-Directed Human Attention

CVPR 2023

Sounak Mondal, Z. Yang, S. Ahn, D. Samaras, G. Zelinsky, M. Hoai

Personalized Image Descriptions from Attention Sequences	CVPR 2026
R. Xue, J. Xu, H. Le, Sounak Mondal , A. Leite, G. Zelinsky, M. Hoai, D. Samaras	
Few-shot Personalized Scanpath Prediction	CVPR 2025
R. Xue, J. Xu, Sounak Mondal , H. Le, G. Zelinsky, M. Hoai, D. Samaras	
Unifying Top-down and Bottom-up Scanpath Prediction using Transformers	CVPR 2024
Z. Yang, Sounak Mondal , S. Ahn, R. Xue, G. Zelinsky, M. Hoai, D. Samaras	
Target-absent Human Attention	ECCV 2022
Z. Yang, Sounak Mondal , S. Ahn, G. Zelinsky, M. Hoai, D. Samaras	
Diffusion-Refined VQA Annotations for Semi-Supervised Gaze Following	ECCV 2024
Q. Miao, A. Graikos, J. Zhang, Sounak Mondal , M. Hoai, D. Samaras	
Characterizing Target-absent Human Attention	CVPRW 2022
Y. Chen, Z. Yang, S. Chakraborty, Sounak Mondal , S. Ahn, D. Samaras, M. Hoai, G. Zelinsky	

SELECTED AWARDS

- ICCV 2025 Doctoral Consortium
- CVPR 2023 Travel Grant Award
- ACM Best B.E. Thesis Award, 2017

ACADEMIC & TECHNICAL DETAILS

Graduate Courses: Computer Vision, Natural Language Processing, Robotics, Machine Learning, Database Systems
Languages & Frameworks: Python, PyTorch, TensorFlow, C++, C, Java, Hadoop (familiar), Spark (familiar)
Service: Reviewer for CVPR, ICCV, ECCV, ICLR, NeurIPS, TPAMI